

Retail Product Sales Analysis Using Excel

Data Exploration and Insights

Project Objective:

To explore a retail product sales dataset using Excel functions - SUM, COUNT, AVERAGE, MIN, MAX, IF, SUMIF, COUNTIF, LEFT, RIGHT, and MID - to perform data exploration, classify products, extract text-based insights, and arrive at meaningful business conclusions from the sales data.

Dataset Description:

The dataset is a retail sales dataset containing 34 product records across multiple categories including Electronics, Fashion, Kitchen, Outdoor, and Accessories. Products are sold across 12 countries — US, UK, Canada, India, Australia, Germany, Italy, France, Spain, China, Russia, and Brazil. Each record includes the product name, brand, selling price, quantity sold, and category.

Column Description

Column Name	Description
Product ID	Unique identifier combining date and country code (e.g. 28-JAN-US)
Product Name	Name of the product (e.g. Laptop, Sneakers, Blender)
Brand Name	Brand of the product (e.g. Dell, Nike, Ninja)
Price (\$)	Selling price of the product in US dollars
Quantity	Number of units sold
Category	Product category — Electronics, Fashion, Kitchen, Outdoor, Accessories
Price Range	High Price (≥ 500) or Standard Price (< 500) — derived using IF formula
Day	Day extracted from Product ID using LEFT formula
Country Code	Country code extracted from Product ID using RIGHT formula
Month	Month extracted from Product ID using MID formula

Data Cleaning and Transformation:

The dataset did not contain any missing values. A new column called Price Range was added in Column G using the IF function to classify products as High Price (≥ 500) or Standard Price (< 500). Three additional columns were created by extracting parts of the Product ID - Day (Column H) using LEFT, Country Code (Column I) using RIGHT, and Month (Column J) using MID.

Visualization and Insights Arrived with Screenshot of Visualization Report/Dashboard:

Descriptive Analysis:

The retail dataset contains 34 products with a total price value of \$10,100. The average product price is \$297.06. Products span across 5 categories and 12 countries, showing a diverse retail portfolio.

Diagnostic Analysis:

The lowest priced product is the Adidas T-shirt at \$30, and the most expensive is the Dell Laptop at \$1,000. The Electronics category alone accounts for \$8,050 out of the total \$10,100 revenue - contributing approximately 80% of total sales value.

Predictive Analysis:

Given that Electronics contributes nearly 80% of total revenue, future inventory and sales planning should prioritise this category. The 11 budget products priced below \$100 show potential for high-volume sales if targeted with promotions.

Prescriptive Analysis:

It is recommended to run discount campaigns on the 11 products priced below \$100 to drive volume sales. Stock levels for high-value Electronics such as Laptops (\$1,000) and Smartphones (\$900) should be maintained consistently as they carry the highest individual revenue contribution.

Reference

Fig 1 & Fig 2 — Task 1: SUM Formula =SUM(D2:D35) | COUNT Formula =COUNT(D2:D35)

Q1. What is the total price of all products in the retail dataset?

Answer: Formula: =SUM(D2:D35) | Result: \$10,100

Q2. How many product records exist in the dataset?

Answer: Formula: =COUNT(D2:D35) | Result: 34 products

Data Analytics (DA) — Module 1: Excel Assignment 1 – Data Exploration / Formula

SCREENSHOTS TASK 1 — BASIC DATA EXPLORATION: SUM, COUNT, AVERAGE FIG 1: CELL M3 SELECTED — FORMULA BAR SHOWS =SUM(D2:D35)

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Task 1 – Sum, Count & Average	
Total Price of All Products	10100
Count of Products	34
Average Price	297.0588235

Task 2 – Minimum & Maximum	
Minimum Price	30
Maximum Price	1000

Task 4 – SUMIF & COUNTIF	
Total Price – Electronics Category	8050
Products with Price < \$100	11

FIG 1: CELL M3 SELECTED — FORMULA BAR SHOWS =SUM(D2:D35) | RESULT: 10100

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Task 1 – Sum, Count & Average	
Total Price of All Products	10100
Count of Products	34
Average Price	297.0588235

Task 2 – Minimum & Maximum	
Minimum Price	30
Maximum Price	1000

Task 4 – SUMIF & COUNTIF	
Total Price – Electronics Category	8050
Products with Price < \$100	11

FIG 2: CELL M4 SELECTED — FORMULA BAR SHOWS =COUNT(D2:D35) | RESULT: 34

Fig 3 & Fig 4 — Task 1: AVERAGE Formula =AVERAGE(D2:D35) | Task 2: MIN Formula =MIN(D2:D35)

Q3. What is the average selling price across all retail products?

Answer: Formula: =AVERAGE(D2:D35) | Result: \$297.06

Q4. What is the lowest priced product in the retail dataset?

Answer: Formula: =MIN(D2:D35) | Result: \$30 (T-shirt by Adidas)

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	35	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	80	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Pebble	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	26	CA	SEP

FIG 3: CELL M5 SELECTED — FORMULA BAR SHOWS =AVERAGE(D2:D35) | RESULT: 297.06

TASK 2 — MINIMUM & MAXIMUM VALUES: MIN, MAX

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	35	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	80	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Pebble	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	26	CA	SEP

FIG 4: CELL M9 SELECTED — FORMULA BAR SHOWS =MIN(D2:D35) | RESULT: 30

Fig 5 & Fig 6 — Task 2: MAX Formula =MAX(D2:D35) | Task 3: IF Formula Applied to Column G

Q5. What is the highest priced product in the retail dataset?

Answer: Formula: =MAX(D2:D35) | Result: \$1,000 (Laptop by Dell)

Q6. Classify each product as "High Price" if price >= \$500, otherwise "Standard Price".

Answer: Formula: =IF(D2>=500,"High Price","Standard Price") | Applied to G2:G35

The screenshot shows an Excel spreadsheet with a retail dataset. The formula bar at the top displays the formula =MAX(D2:D35) and the result 1000. The dataset includes columns for Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The results of the MAX formula are shown in the summary table on the right.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	160	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
05-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	05	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 5: CELL M10 SELECTED — FORMULA BAR SHOWS =MAX(D2:D35) | RESULT: 1000

Task 3 — Logical Function: IF (Price Range Column)

The screenshot shows the same Excel spreadsheet as Figure 5, but with the IF formula applied to column G. The formula bar at the top displays the formula =IF(D2>=500,"High Price","Standard Price"). The results of the IF formula are shown in the Price Range column of the dataset.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	160	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
05-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	05	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 6: CELL G2 SELECTED — FORMULA BAR SHOWS =IF(D2>=500,"HIGH PRICE","STANDARD PRICE")
| APPLIED TO G2:G35

Task 4 — Conditional Functions: SUMIF & COUNTIF

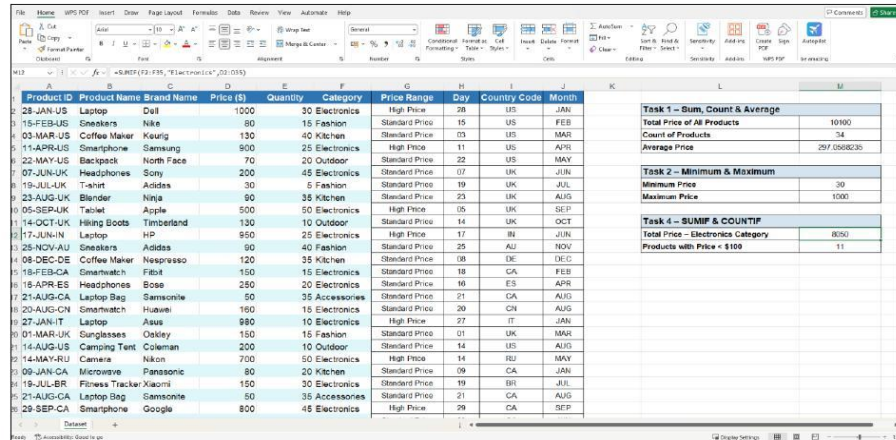
Fig 7 & Fig 8 — Task 4: SUMIF Formula | COUNTIF Formula =COUNTIF(D2:D35,"<100")

Q7. What is the total revenue contributed by the Electronics category alone?

Answer: Formula: =SUMIF(F2:F35,"Electronics",D2:D35) | Result: \$8,050

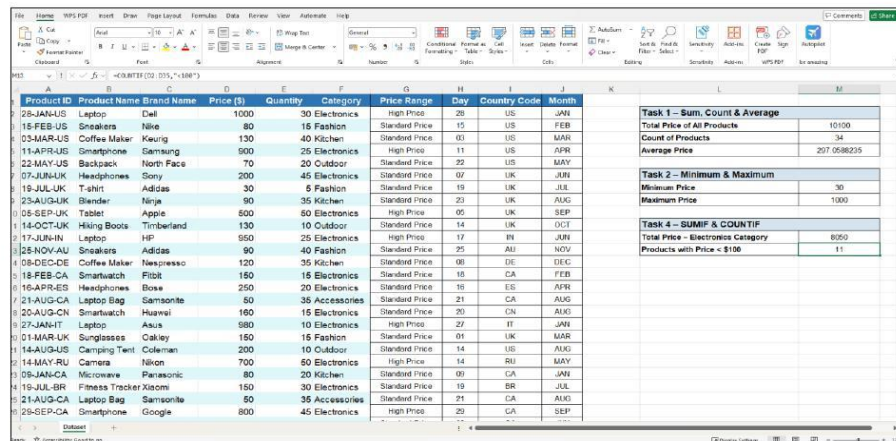
Q8. How many products in the dataset are priced below \$100?

Answer: Formula: =COUNTIF(D2:D35,"<100") | Result: 11 products



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	60	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 7: CELL M13 SELECTED — FORMULA BAR SHOWS =SUMIF(F2:F35,"ELECTRONICS",D2:D35) | RESULT: 8050



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	60	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 8: CELL M14 SELECTED — FORMULA BAR SHOWS =COUNTIF(D2: D35,"<100") | RESULT: 11

Task 5 — Text Functions: LEFT, RIGHT, MID

Fig 9 & Fig 10 — Task 5: LEFT Formula =LEFT(A2,2) | RIGHT Formula =RIGHT(A2,2)

Q9. Extract the sale day from the Product ID field into a separate column.

Answer: Formula: =LEFT(A2,2) | Extracts first 2 characters (Day) | Applied to H2:H35

Q10. Extract the country code from the Product ID field into a separate column.

Answer: Formula: =RIGHT(A2,2) | Extracts last 2 characters (Country Code) | Applied to I2:I35

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Addidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Addidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fibit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	60	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 9: CELL H2 SELECTED — FORMULA BAR SHOWS =LEFT(A2,2) | EXTRACTS DAY (FIRST 2 CHARS) | APPLIED TO H2:H35

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Addidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Addidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fibit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	60	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 10: CELL I2 SELECTED — FORMULA BAR SHOWS =RIGHT(A2,2) | EXTRACTS COUNTRY CODE (LAST 2 CHARS) | APPLIED TO I2:I35

Fig 11 — Task 5: MID Formula =MID(A2,4,3) | Extracts Month from Product ID

Q11. Extract the month of sale from the Product ID field into a separate column.

Answer: Formula: =MID(A2,4,3) | Starts at position 4, takes 3 characters (Month) | Applied to J2:J35

The screenshot shows an Excel spreadsheet with a product inventory table and summary tasks. The formula bar for cell J2 shows the formula =MID(A2,4,3).

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	600	60	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	80	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	160	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	60	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	150	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	60	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Task 1 – Sum, Count & Average

Total Price of All Products	10100
Count of Products	34
Average Price	297.0588235

Task 2 – Minimum & Maximum

Minimum Price	30
Maximum Price	1000

Task 4 – SUMIF & COUNTIF

Total Price – Electronics Category	8050
Products with Price < \$100	11

FIG 11: CELL J2 SELECTED — FORMULA BAR SHOWS =MID(A2,4,3) | EXTRACTS MONTH (CHARS 4-6) | APPLIED TO J2:J35

Conclusion:

This project successfully demonstrated the use of core Excel functions to explore and analyse a retail product sales dataset. All five tasks were completed — basic aggregation using SUM, COUNT, and AVERAGE; range identification using MIN and MAX; logical classification using IF; conditional analysis using SUMIF and COUNTIF; and text extraction using LEFT, RIGHT, and MID. The analysis confirmed that Electronics is the dominant revenue category contributing \$8,050 out of \$10,100 total, and that 11 budget-priced products under \$100 represent an opportunity for volume-driven sales strategies across the 12 countries covered in the dataset.